

Visualizing Lattice-based Post-Quantum Cryptographic Algorithms

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Abstract. This paper presents the development of a visualization tool named (final name later) The abstract should briefly summarize the contents of the paper in 150–250 words.

Keywords: Cryptography · Post-Quantum · Another keyword.

1 Introduction

Cryptography is the study of techniques and protocols relating to securing communications (Rivest, 1990). Typically, this consists of constructing and analyzing protocols that prevent outside parties from reading private messages.

Prior to the modern era, most of cryptography was encryption, which is a process that transforms some form of information (plaintext) into an alternative form that only certain parties can decode (ciphertext). What is now known as classical cryptography consisted of using ciphers that could be practically computed and solved by hand. However, after devices such as the Enigma machine and other computational devices were developed, these ciphers became trivially easy to break.

Now, history repeats itself with the development of quantum computers as they threaten to render multiple modern cryptographic algorithms useless. Quantum algorithms such as the ones proposed by Peter Shor are theorized to be able to break protocols like RSA given a strong enough quantum computer (Shor, 1994). Because of this, cryptology experts around the world have looked into new protocols that might have the potential to defend against them.

NIST (National Institute of Standards and Technology) is one of the organizations at the forefront of the effort. In 2016, they put out a call for proposals that might be used in post-quantum cryptography. Eight years later, they released three standards based on those submissions, two of which are lattice-based; ML-KEM and ML-DSA, based on the Kyber and Dilithium algorithms respectively. ML-KEM is used for key encapsulation, while ML-DSA is used for digital signatures.

Despite their importance, there are a limited number of resources available for learning about them. In some part this is because they are relatively new

standards. However, it can also be attributed to their complexity and nature as lattice-based. Aside from NIST’s released papers (NIST, 2024), attempts have been made to visualize the scheme and other lattice-based algorithms, including the following:

- pqc-vizz (ML-KEM, ML-DSA, SLH-DSA)
- CrypTool 2 (cryptography module)

However, they tend to skip parts of the step-by-step process, not show the background calculations, as well as simplify the lattice too much (Rodríguez et al., 2024; Kalava et al., 2025).

Thus, this work aims to develop a visualization tool that allows users to view the complete process behind ML-KEM and ML-DSA.

2 Related Works

TODO: The idea of using visualization tools to facilitate the understanding of complex topics is not new. Visualization tools for topics such as . However, the current tools available for visualizing ML-KEM and ML-DSA are

2.1 ML-KEM

Explain ML-KEM and how it works

2.2 ML-DSA

Explain ML-KEM and how it works

2.3 Lattice Visualization

Explain why lattice visualizations are hard

2.4 Existing Visualizations

Examples of existing visualizations and pros and cons

3 Visualization Tool Design

The developed visualization tool is a web application that provides visualizations for ML-KEM, ML-DSA, LLL, and BKZ. (elaborate later)

3.1 Architecture

The visualization tool is built using Next.js, a React-based framework, for the frontend, and primarily Typescript for the backend processes. The ML-KEM and ML-DSA algorithms are implemented using an existing Javascript library and imported as needed. (insert diagram later + be more thorough)

3.2 ML-KEM

This part is for explaining how ML-KEM is implemented/how the code works

3.3 ML-DSA

This part is for explaining how ML-DSA is implemented/how the code works

4 Results?

5 Conclusion

Theorem 1. *This is a sample theorem. The run-in heading is set in bold, while the following text appears in italics. Definitions, lemmas, propositions, and corollaries are styled the same way.*

Proof. Proofs, examples, and remarks have the initial word in italics, while the following text appears in normal font.

For citations of references, we prefer the use of square brackets and consecutive numbers. Citations using labels or the author/year convention are also acceptable. The following bibliography provides a sample reference list with entries for journal articles [1], an LNCS chapter [2], a book [3], proceedings without editors [4], and a homepage [5]. Multiple citations are grouped [1–3], [1, 3–5].

Acknowledgments. A bold run-in heading in small font size at the end of the paper is used for general acknowledgments, for example: This study was funded by X (grant number Y).

Disclosure of Interests. It is now necessary to declare any competing interests or to specifically state that the authors have no competing interests. Please place the statement with a bold run-in heading in small font size beneath the (optional) acknowledgments³, for example: The authors have no competing interests to declare that are relevant to the content of this article. Or: Author A has received research grants from Company W. Author B has received a speaker honorarium from Company X and owns stock in Company Y. Author C is a member of committee Z.

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